

GeoAI Deforestation Risk Intelligence Platform

2026

An AI-Powered Spatial Decision-Support Platform for Deforestation Risk Assessment in Rondônia, Brazil

PRINCIPAL DEVELOPER

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PRODUCTION PLATFORM

amazon-deforestation-risk-3d.vercel.app

STUDY GEOGRAPHY

Rondônia, Brazilian Amazon Basin

REPOSITORY

github.com/mamdouhahmed5455-lang/Amoazon2

CORE DECISION-SUPPORT PURPOSE

"An AI-assisted spatial decision-support prototype for interpreting and prioritizing relative deforestation risk."

STUDY GEOGRAPHY

Rondônia, Brazil

Legal Amazon Basin

SPATIAL COVERAGE

~150,000 Cells

Precomputed Grid

MODEL PIPELINE

Offline XGBoost

Gradient Boosted Trees

TRACK 1
BENCHMARK

ROC-AUC 0.82

F1-Score: 0.81

TRACK 2 RESEARCH

ROC-AUC 0.7474

5.0× Retrospective Top-Decile Enrichment

WHAT

An end-to-end GeoAI decision-support prototype that synthesizes multi-source geospatial and demographic indicators into interpretable, relative spatial risk rankings.

WHERE

State of Rondônia, Brazilian Amazon Basin — covering approximately 150,000 precomputed spatial cells, contextualized by 25 years of PRODES deforestation history (2001–2025) across all 9 Legal Amazon states.

TECH

Offline-trained XGBoost classification pipeline, spatial feature engineering (STRtree / cKDTree / Copernicus DEM), and a static client-side WebGL (Deck.gl / Mapbox GL JS) visualization engine deployed on Vercel Edge CDN with zero backend database requirement.

DEMONSTRATED

Two Distinct Evidence Tracks: Documented Production Benchmark (Track 1: ROC-AUC 0.82, Precision 0.79, Recall 0.84) + Independent Research Evaluation (Track 2: ROC-AUC 0.7474, 95% CI: [0.5837, 0.9036], Brier 0.00693, 5.0× Retrospective Top-Decile Enrichment).

WHY

The project explores how environmental observations can be organized into a structured workflow for spatial risk screening, prioritization, interpretation, and verification planning.

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Executive Summary

The **GeoAI Deforestation Risk Intelligence Platform** is an end-to-end GeoAI decision-support system designed to assist environmental analysts in spatially screening, prioritizing, and interpreting deforestation risk across the Brazilian Amazon.

The platform synthesizes multi-source geospatial and demographic data into a precomputed spatial risk dataset covering approximately **150,000 spatial cells** in the state of **Rondônia, Brazil**. An offline-trained **XGBoost** classification pipeline underpins the risk scoring. Risk outputs represent **relative spatial risk rankings**, not calibrated probabilities or forecasts of deforestation occurrence.

The platform is delivered as a **browser-based static web application** hosted on Vercel, requiring no server-side database, enabling immediate interactive exploration via WebGL-powered 2D/3D visualization.

Scientific evidence is maintained across two explicitly separated tracks:

TRACK 1 — DOCUMENTED PRODUCTION BENCHMARK				TRACK 2 — INDEPENDENT RESEARCH MODEL V2	
ROC-AUC	PRECISION	RECALL	F1-SCORE	ROC-AUC	0.7474
0.82	0.79	0.84	0.81	95% CI	[0.5837, 0.9036]
				PR-AUC	0.005534
				Brier Score	0.00693
				Top-Decile Lift	5.0×

A **25-year PRODES-based historical intelligence** layer (2001–2025) across all nine states of the Brazilian Legal Amazon provides contextual depth for spatial interpretation.

Integrity Statement: This platform does not claim real-time prediction, live satellite monitoring, causal inference, measured financial ROI, measured carbon savings, autonomous enforcement, or proven operational deployment. All risk outputs are heuristic spatial signals for decision-support purposes only.

Context

The Amazon rainforest is Earth's largest terrestrial carbon reservoir and biodiversity habitat, yet it faces persistent fragmentation from infrastructure expansion, agricultural encroachment, and illegal clearing activity.

Brazil operates world-class forest monitoring systems:

- **PRODES** (INPE): annual consolidated mapping of deforestation polygons ≥ 6.25 ha
- **DETER** (INPE): rapid optical and radar alert stream detecting active clearings ≥ 3 ha
- **Global Forest Watch / GLAD**: global canopy disturbance alerts

These systems represent significant achievements in environmental surveillance. The challenge they face is not a failure of data or technology.

The Gap This Project Addresses

"The challenge is not the absence of forest-monitoring data, but the gap between environmental observations and an integrated workflow for spatial risk screening, prioritization, interpretation, and verification planning."

Environmental protection agencies operate under severe logistical, personnel, and vehicle constraints across millions of square kilometers of remote territory. Even with alert data available, enforcement coordinators require:

WORKFLOW NEED	DESCRIPTION
Spatial prioritization	Which zones accumulate the highest structural pressure across multiple risk factors simultaneously?
Integrated explanation	What combination of anthropogenic drivers underlies flagged cells?
Contextual history	How does current spatial risk relate to 25 years of observed deforestation patterns?
Sensitivity exploration	How might risk distributions shift under alternative infrastructure or policy conditions?
Verification planning	Which areas warrant ground-truthing or targeted monitoring attention?

This platform is designed as a **complementary AI-assisted decision-support layer** to help bridge that workflow gap — operating alongside, not replacing, authoritative monitoring infrastructure.

CORE PRINCIPLE

"A complementary intelligence layer, not a replacement for monitoring systems."

The platform delivers a structured decision-support workflow, from multi-source data ingestion through precomputed spatial inference to interactive browser-based analyst tools:

DECISION-SUPPORT PIPELINE

MULTI-SOURCE DATA

Road Network · Historical Forest Loss · Population Pressure · Elevation Constraints



OFFLINE FEATURE ENGINEERING (PYTHON)

STRtree / cKDTree spatial joins · PRODES WFS · OSM Overpass · IBGE Census · Copernicus DEM



XGBOOST RISK SCORING (OFFLINE PIPELINE)

Precomputed spatial risk scores for ~150,000 cells · Out-of-time temporal partitions



BROWSER APPLICATION (STATIC / WEBGL)

Risk Signal → Spatial Explanation → Prioritization → Historical Context → Scenario
Sensitivity → Verification Planning

The system is designed so that a trained environmental analyst can navigate from a raw spatial risk signal through feature-level explanation, historical context, sensitivity exploration, and structured verification planning — entirely within the browser, without requiring GIS desktop software or specialist data engineering capacity.

5.1 AI-Assisted Spatial Risk Ranking

The platform integrates AI-assisted spatial risk ranking **alongside** historical observation and monitoring data. Rather than replacing authoritative detection systems, it provides a complementary structural risk layer at the multi-factor intersection of road proximity, historical forest loss, population pressure, and terrain accessibility.

Risk outputs are explicitly described as **relative spatial risk scores** — heuristic signals derived from the learned relationship between spatial predictors and historical deforestation outcomes. They are not calibrated probabilities or forecasts.

5.2 Explainability Through Dual-Track Evidence

Explainability is provided through two scientifically distinct methods:

- **Track 1:** Documented global feature-importance evidence (analytical evidence), establishing the directional role of each predictor in the production benchmark
- **Track 2:** TreeSHAP attribution applied specifically to the independent Research Model V2, providing cell-level quantitative driver decomposition

Important: TreeSHAP attribution belongs to the Track 2 Research Model V2. It is not claimed as the explainability method for the Track 1 production benchmark.

5.3 Client-Side Scenario Sensitivity Simulation

The platform includes a browser-embedded **parametric sensitivity simulation** module. Analysts can adjust road expansion, population pressure, fire frequency, and policy enforcement parameters to observe how the risk distribution responds under alternative conditions.

This is **exploratory sensitivity simulation, not validated forecasting**. Results are illustrative policy-sensitivity signals, not predictive policy forecasts.

5.4 Interactive WebGL-Based Geospatial Exploration

The browser application renders approximately 150,000 spatial cells using WebGL through Deck.gl and Mapbox GL JS, supporting 3D column extrusion, 2D orthographic mapping, density heatmap rendering, and spatial hotspot filtering. Client-side rendering is optimized for interactive exploration of the precomputed spatial dataset.

5.5 Dual-Track Scientific Transparency

The platform explicitly separates the Track 1 documented production benchmark from the Track 2 independent research audit, with metrics, methodology, limitations, and evidence levels clearly documented for both. This architecture demonstrates scientific maturity rather than inflating a single performance figure.

The platform maintains a strict separation between the **offline research and training environment** and the **production browser experience**:

FULL SYSTEM ARCHITECTURE

RESEARCH & TRAINING ENVIRONMENT (OFFLINE PYTHON)

- DATA SOURCES: INPE PRODES WFS · OpenStreetMap (Overpass) · IBGE Census · Copernicus DEM GLO-90m
- FEATURE ENGINEERING: STRtree Spatial Indexing · cKDTree Spatial Join · Road proximity · Historical loss · Population · Elevation
- XGBOOST PIPELINE: Gradient Boosted Decision Trees (Log-Loss) · Out-of-time partitions (2022/2023/2024) · Geographic latitude holdout (~11.0°) · 1,000-iteration bootstrap CI
- OUTPUT: Precomputed spatial inference · ~150,000 cells · Risk scores · Feature values

↓ Static asset deployment

PRODUCTION BROWSER EXPERIENCE (STATIC WEB APPLICATION)

- Deck.gl WebGL Engine + Mapbox GL JS
- Vanilla JavaScript (ES6 modular) · Chart.js · D3.js
- Interactive 2D/3D Risk Maps · Decision Story · Scenario Sensitivity · History Layer
- Zero server-side database · Vercel Edge CDN

Key separation principle: No live model training or real-time prediction occurs in the browser. The browser application loads precomputed spatial data and delivers interactive visualization, explanation, and scenario sensitivity tools.

Data Engineering

FEATURE	SOURCE	ENGINEERING METHOD
Road Proximity	OpenStreetMap (Overpass API)	STRtree spatial indexing + cKDTree distance to nearest highway
Historical Deforestation	INPE PRODES WFS (2021–2024)	Annual clear-cut polygon intersection per spatial cell
Population Pressure	IBGE 2022 Decennial Census	Population density proxy spatially joined to grid centroids
Elevation Constraints	Copernicus DEM GLO-90m	Terrain elevation values per cell centroid

Model Pipeline

- **Algorithm:** XGBoost gradient-boosted decision trees with log-loss objective
- **Temporal partitioning:** Temporal target-year isolation was applied in the Model V2 evaluation protocol (2022 training, 2023 validation, 2024 test), with predictor timing documented per source. Temporal alignment is not identical across all source variables and evaluation periods; therefore, no blanket zero-leakage claim is made.
- **Geographic holdout:** North/south split at -11.0° latitude for spatial generalization testing
- **Confidence intervals:** 1,000-iteration bootstrap resampling on the 2024 held-out partition

Browser Application

MODULE	DESCRIPTION
Interactive Risk Map (index.html)	3D extrusion / 2D orthographic / heatmap / hotspot views across ~150,000 spatial cells
Decision Story	5-step guided analytical narrative embedded in the dashboard
Historical Intelligence	25-year PRODES time series (2001–2025), 9 Legal Amazon states
Scenario Simulator	Client-side parametric sensitivity analysis
Spatial Analysis Workspace	Cluster analysis, corridor convergence, spatial filtering
AI Model Diagnostics	Feature importance, confusion matrix, model architecture visualization

RIGOR PRINCIPLE

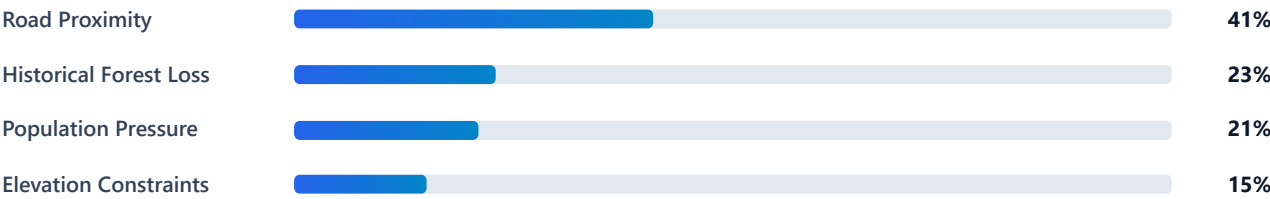
"Two evidence tracks, clearly separated: documented production benchmark vs. independent research evaluation."

TRACK 1 — DOCUMENTED PRODUCTION BENCHMARK (CANONICAL REFERENCE)	
Model	XGBoost (Baseline)
ROC-AUC	0.82
Precision	0.79
Recall	0.84
F1-Score	0.81
95% Bootstrap CI	Not documented upstream
Evidence Level	Canonical Benchmark (Locked)
Explainability	Global feature-importance evidence

TRACK 2 — INDEPENDENT RESEARCH MODEL V2	
Model	XGBoost v2.0.0
ROC-AUC	0.7474
95% Bootstrap CI	[0.5837, 0.9036]
PR-AUC	0.005534
Brier Score	0.00693
Test Events (2024)	10 observed positives
Top-Decile Lift	5.0× retrospective
Evidence Level	Moderate
Explainability	TreeSHAP attribution

Explicit Scientific Statement: Track 2 (Model V2) is an independent empirical verification research experiment. It is not presented as a reproduction of the Track 1 benchmark and does not replace it.

Feature-Importance Evidence (Track 1 — Documented Global Attribution)



These are **documented global feature-importance values** (analytical evidence). They are **not** XGBoost model coefficients and should not be interpreted as causal effect sizes.

Track 2 Critical Interpretation Constraints

Risk Score Interpretation: Brier score alone does not establish probabilistic calibration. Model V2 outputs must be interpreted as **relative spatial risk rankings**, not calibrated probabilities or frequentist estimates of deforestation occurrence likelihood.

BOUNDARY PRINCIPLE

"Relative risk ranking, not calibrated probability."

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Historical Environmental Intelligence

The platform integrates a **25-year historical environmental intelligence layer** based on Brazil's official PRODES deforestation accounting program.

DIMENSION	DETAIL
Data source	INPE PRODES (TerraBrasilis WFS)
Temporal range	2001–2025
Geographic scope	All 9 states of the Brazilian Legal Amazon
Focus state	Rondônia
Polygon threshold	≥ 6.25 hectares per clear-cut unit

Historical intelligence serves three decision-support functions:

1. **Contextual anchoring:** Allows analysts to situate current risk signals within long-term deforestation trajectories per state and sub-region
2. **Baseline reference:** Establishes empirical deforestation patterns used as training labels in the XGBoost pipeline
3. **State comparison:** Enables cross-state comparison of annual deforestation rates across all nine Legal Amazon states from 2001 to 2025

Important distinction: Historical intelligence reflects **retrospective measurement** of past deforestation events. It does not constitute forward-looking prediction of future deforestation occurrence.

WORKFLOW PRINCIPLE*"From risk signal to verification planning."*

The platform guides analysts through a structured five-step decision-support workflow embedded in the dashboard interface:

1

Where is the risk?

Identify spatial cells with the highest relative risk scores across Rondônia using the interactive 3D / 2D map interface.

2

Why is it flagged?

Inspect the feature-level explanation for a selected cell: road proximity, historical forest loss, population pressure, elevation. Understand the multi-factor combination underlying the risk signal.

3

How does the spatial pattern change?

Use the spatial analysis workspace to examine cluster distributions, road corridor convergence, and risk tier transitions across adjacent cells.

4

What should be investigated next?

Apply the scenario sensitivity simulator to understand how the spatial risk distribution responds to parametric changes in key pressure variables.

5

What additional verification or analysis may be valuable?

Use prioritized risk rankings to structure verification planning. All planning windows and inspection guidance are **illustrative and non-prescriptive**.

Scope boundary: The workflow supports **strategic screening and prioritization**. It does not autonomously dispatch enforcement resources, generate legal instruments, or replace ground-truth verification.

11 Scenario Sensitivity Analysis

The platform includes a client-side parametric sensitivity simulation module. Analysts can interactively adjust four pressure variables:

VARIABLE	DESCRIPTION
Road Expansion	Simulates the effect of additional road corridor proximity on spatial risk distribution
Population Pressure	Models changes in demographic density and its influence on risk signal
Fire Frequency	Adjusts fire-related pressure signals across the spatial grid
Protection Policy	Toggles the effect of simulated reserve enforcement constraints on risk tier allocation

Scientific Classification: This is **exploratory sensitivity simulation, not validated forecasting**. Results are illustrative sensitivity signals for qualitative scenario reasoning. The simulation does not constitute a validated policy forecast, represent a predictive model of future deforestation rates, or claim causal inference between policy interventions and outcomes.

12 Technical Feasibility

The platform is a **fully functional static web prototype** demonstrating end-to-end feasibility of the GeoAI decision-support concept:

COMPONENT	FEASIBILITY BASIS
Offline Python pipeline	Fully implemented and documented in repository
Precomputed spatial dataset	Generated and deployed as static asset (~150,000 cells)
XGBoost model pipeline	Implemented with documented evaluation protocol
Deck.gl / Mapbox visualization	Operational in production deployment
Client-side sensitivity simulation	Functional parametric simulation module
Vercel deployment	Live at canonical production URL

Client-side rendering is optimized for interactive exploration of the precomputed spatial dataset. No unsupported latency benchmarks are claimed.

A transition from prototype to operational-scale deployment would require automated data ingestion pipelines, scheduled batch model re-scoring, tiled spatial data delivery, and role-based access control for institutional analyst workflows.

The platform operates as an analyst-facing **browser-based decision-support tool** requiring no specialized GIS software or server-side infrastructure management.

Target Analyst Profile

- Environmental intelligence analysts reviewing spatial pressure patterns
- Forest protection planners prioritizing patrol allocation under logistical constraints
- Conservation researchers exploring multi-factor deforestation risk drivers

Potential Future Institutional Integration

Potential future integration with environmental monitoring and conservation institutions — such as IBAMA, ICMBio, or SEDAM-RO — is subject to institutional validation, data governance requirements, and formal deployment agreements. **No current partnership, institutional deployment, or government adoption is claimed.**

Planning Windows and Inspection Guidance

Any inspection prioritization windows or planning guidance presented in the platform are:

- **Illustrative:** provided as suggested planning guidance for demonstration purposes only
- **Non-prescriptive:** not to be interpreted as operational SLAs, guaranteed response times, or official operating procedures

Scope: This section discusses economic feasibility qualitatively only. No invented currency figures, validated ROI claims, or financial savings projections are presented. All economic benefits remain qualitative hypotheses requiring field pilot validation.

Current Prototype Cost Profile

COST DIMENSION	CURRENT STATUS
Data Acquisition	PRODES/IBGE API access (open), OSM highway data, Copernicus DEM
Compute (Research)	Python environment, XGBoost training, bootstrap resampling
Infrastructure	Vercel Edge hosting (static assets); negligible at current scale
Maintenance	Dataset refresh cadence, model re-evaluation, browser compatibility

Production-Scale Cost Categories

CATEGORY	DESCRIPTION
Automated Data Pipelines	Scheduled satellite data ingestion and feature re-computation
Batch Model Inference	Periodic re-scoring of spatial grid as new deforestation polygons are registered
Cloud Tile Caching	Tiled spatial delivery infrastructure for larger geographic extents
Analyst Training	Human capacity development for platform interpretation and workflow integration
Institutional Integration	IT infrastructure, data governance, access control
Field Verification	Ground-truth validation costs for priority zone inspection

A complementary spatial screening layer has the potential — subject to field pilot validation — to reduce the investigative surface area that analysts must manually prioritize. No measured efficiency gains, patrol cost reductions, carbon savings, or financial ROI are claimed.

The platform is positioned as a **complementary AI-assisted decision-support layer**, not a replacement for or competitor to authoritative monitoring systems.

SYSTEM	AGENCY	PRIMARY SCOPE	CADENCE	RELATIONSHIP
PRODES	INPE (Brazil)	Annual official consolidated deforestation accounting	Annual	Complementary PRODES maps confirmed historical loss; this platform provides precomputed spatial risk screening alongside PRODES-derived intelligence
DETER	INPE (Brazil)	Rapid optical/radar alert stream ≥ 3 ha	Daily/Weekly	Complementary DETER detects active clearings; this platform models underlying multi-factor structural susceptibility
Global Forest Watch	WRI	Global canopy disturbance alerts	Near-continuous	Geographic complement GFW operates globally; this platform provides localized Amazonian anthropogenic risk modeling
MapBiomas	NGO Consortium	Multi-decade land cover classification	Annual	Temporal complement MapBiomas provides historical land use; this platform focuses on current structural risk ranking
Planet NICFI	Norway / Planet	High-resolution monthly basemaps	Monthly	Layer complement NICFI provides visual imagery; this platform provides ML risk inference on demographic and infrastructure drivers

Positioning niche: An explainable spatial screening layer bridging environmental observation data and structured verification planning workflows. This platform does not claim superiority over any authoritative monitoring system.

Scientific transparency is a **core design principle** of this project, not an afterthought. The following limitations are disclosed explicitly to demonstrate scientific maturity.

- 1 **Risk Score Interpretation** — Displayed risk scores are relative spatial risk rankings, not calibrated probabilities. They should not be interpreted as calibrated probabilities of deforestation occurrence, likelihood percentages of clearing events, or actuarial estimates of future forest loss.
- 2 **Track 2 Small Sample Constraint** — The Track 2 held-out 2024 test partition contains only **10 observed positive deforestation events** out of 7,045 spatial cells (0.142% prevalence). Confidence intervals are wide: ROC-AUC 95% CI [0.5837, 0.9036]. Point estimates carry substantial finite-sample uncertainty.
- 3 **Probabilistic Calibration** — The Brier score (0.00693) alone does not establish probabilistic calibration. A low Brier score in severely imbalanced datasets primarily reflects the class distribution, not calibration quality.
- 4 **Temporal Demographic Alignment** — Population feature temporal alignment is not perfectly consistent: 2022 and 2023 partitions use IBGE Census data published after the respective prediction cutoffs (static proxies). Universal zero temporal leakage is not claimed.
- 5 **Scenario Simulation Scope** — The scenario sensitivity module is exploratory parametric simulation, not validated forecasting. It does not constitute a calibrated policy prediction model and does not claim causal inference.
- 6 **Non-Causal Inference** — Feature importance evidence and SHAP attributions reflect predictive correlations within the model, not real-world causal treatment effects. No causal inference is claimed.
- 7 **Track 1 Reproduction** — Track 1 exact reproduction is not independently claimed when original training artifacts from the documented benchmark are unavailable. Track 1 metrics are the canonical documented reference values.
- 8 **Precomputed Dataset** — The production browser application loads a static precomputed spatial dataset. No live model inference, real-time satellite data integration, or dynamic score updating occurs in the browser.
- 9 **Heuristic UI Layers** — Some dashboard layers (e.g., risk tier thresholds, planning guidance text) incorporate heuristic classification decisions. These are clearly documented and do not claim statistical derivation.

PHASE 1**Prototype Consolidation**

Cross-browser compatibility testing, performance profiling, accessibility review, reproducibility packaging.

PHASE 2**Automated Data Pipelines**

Scheduled PRODES/OSM/IBGE ingestion, DEM refresh pipelines, temporal alignment tracking.

PHASE 3**Operational Model Serving**

Cloud-scheduled batch re-scoring, model versioning, automated evaluation against new held-out partitions.

PHASE 4**Multi-Source Validation**

Benchmarking against DETER alerts, cross-referencing GLAD/RADD, field verification logistics, independent audit.

PHASE 5**Regional Expansion**

Extension beyond Rondônia to additional Legal Amazon states, cross-state feature standardization.

PHASE 6**Institutional Integration**

Formal engagement with potential conservation institutions, data governance, GPS/GeoPackage field export (QField, Garmin GPX).

All future development phases are **aspirational roadmap stages**, subject to resource availability and institutional requirements. No current partnerships or deployment commitments are implied.

The platform's architectural approach — precomputed spatial risk scoring, WebGL-based interactive visualization, multi-factor feature engineering, and dual-track scientific transparency — **could potentially evolve** beyond the current Rondônia prototype.

Possible Future Directions

- **Pan-Amazonian expansion:** Extension across the full Brazilian Legal Amazon and potentially adjacent Amazonian nations (Bolivia, Peru, Colombia, Venezuela) where analogous spatial predictors apply
- **Other tropical forest biomes:** Adaptation to Congo Basin, Southeast Asian peatlands, or other regions facing analogous anthropogenic deforestation pressure patterns
- **Multi-hazard environmental intelligence:** Integration of wildfire risk, habitat fragmentation metrics, or hydrological stress indicators alongside deforestation risk
- **Interoperability with international monitoring frameworks:** Data exchange protocols with GFW, GLAD/RADD, or INPE alert infrastructure for cross-platform verification workflows

The current system operates exclusively within Rondônia, Brazil, as a browser-based prototype with a precomputed static dataset. Long-term global potential is a **future vision** contingent on additional research, validation, resource mobilization, and institutional collaboration. **No current global operational capability is claimed.**

VISION PRINCIPLE

"Prototype today, scalable environmental intelligence tomorrow."

The **GeoAI Deforestation Risk Intelligence Platform** demonstrates that rigorous scientific discipline, careful engineering execution, and responsible AI practice can coexist with visual excellence and practical decision-support utility.

Architectural Progression: From Prototype to Future Environmental Intelligence

PHASE 1 · PROTOTYPE

Current Implementation

High-performance static browser application covering ~150,000 precomputed cells in Rondônia, delivering WebGL 2D/3D visualization with zero server database requirement.

PHASE 2 · EVIDENCE

Dual-Track Rigor

Transparent empirical validation separating documented production benchmarks (Track 1: 0.82) from out-of-time retrospective research evaluation (Track 2: 0.7474 with 5.0× lift).

PHASE 3 · DECISION

Practical Workflow

Structured 5-step operational protocol: spatial risk screening → multi-factor driver inspection → spatial cluster analysis → parametric sensitivity simulation → verification planning.

PHASE 4 · VISION

Future Infrastructure

Methodological architecture positioned for potential multi-biome expansion, multi-hazard environmental risk layering, and sovereign institutional interoperability (pending resources).

SCIENTIFIC RIGOR

Dual-track evidence architecture with explicit benchmark separation, bootstrapped confidence intervals, and comprehensive limitation disclosure.

ENGINEERING EXECUTION

End-to-end GeoAI prototype from offline data engineering through precomputed spatial inference to WebGL browser visualization — fully operational at the production URL.

RESPONSIBLE AI

Explicit rejection of unsupported probability claims, causal inference assertions, and promotional exaggeration. Honest, interpretable spatial risk ranking.

ENVIRONMENTAL RELEVANCE

A complementary decision-support layer targeting the workflow gap between environmental observation data and structured spatial prioritization.

The platform successfully provides a complementary spatial screening tier that translates multi-source geospatial data into prioritized, interpretable **relative risk rankings** across approximately **150,000 spatial cells** in Rondônia, Brazil, supported by dual-track scientific benchmarks, a 25-year historical intelligence foundation, and high-performance WebGL visualization.

PRODUCTION PLATFORM

<https://amazon-deforestation-risk-3d.vercel.app/>

GITHUB REPOSITORY

<https://github.com/mamdouhahmed5455-lang/Amazon2>

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All metrics, claims, and technical descriptions reflect the current state of the repository. No additional claims, institutional partnerships, or operational deployments are implied beyond those explicitly documented herein.